

Synthetic Data Generation for Healthcare

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Abstract—Synthetic healthcare data generation has emerged as a critical approach to addressing privacy constraints that limit the availability of real patient records for machine learning research. This paper presents a comprehensive evaluation of five generative modelling architectures applied to the task of generating realistic first-day ICU lab profiles from the MIMIC-III clinical database: a standard Conditional Variational Autoencoder (cVAE), a Wasserstein Autoencoder with Sliced Wasserstein Distance regularization (WAE-SWD), a Generative Adversarial Network (GAN), an Autoregressive LSTM (ARM-LSTM) [9], and a Probabilistic Conditional Autoregressive Model (PCARM). All models were conditioned on patient demographics, ICU care unit, length of stay, primary diagnosis group, and 31 Elixhauser comorbidity flags, and evaluated against 23 first-day lab values spanning core chemistry, blood gas, haematology, coagulation, and liver panels using the Kolmogorov-Smirnov statistic as the primary distributional fidelity metric. We report that WAE-SWD and PCARM emerge as the strongest performing architectures with mean KS statistics [7] of 0.142 and 0.153 respectively. The standard cVAE provides a reliable baseline at mean KS 0.183, while the GAN exhibits high variance across labs making it unsuitable for clinical applications despite isolated strong results. ARM-LSTM proves fundamentally unsuitable for static lab profile generation, achieving KS=1.0 on six labs due to architectural mismatch with the non-sequential nature of the task. A central finding of this work is that data pipeline quality had a greater impact on model performance than architectural choice.

I. INTRODUCTION

Healthcare data is among the most valuable and most restricted data in existence. Strict privacy regulations such as HIPAA limit the sharing of patient records, creating a significant bottleneck for researchers, data scientists, and machine learning practitioners who need large, diverse datasets to develop and validate clinical models.

In this study, we evaluate the efficacy of different machine learning architectures in generating high-fidelity synthetic

healthcare data using the MIMIC-III database. While synthetic data is essential for preserving patient privacy in research, capturing the complex, non-linear dependencies of clinical measurements remains a significant challenge. Our project establishes a unified preprocessing framework to compare multiple models: Conditional Variational Autoencoder (cVAE), Autoregressive Model (ARM), Generative Adversarial Network (GAN), Probabilistic Conditional Autoregressive Model (PCARM), and Recurrent Neural Network (RNN).

For comparison, we developed a standardized pipeline for extracting and cleaning a 23-lab panel from the MIMIC-III LABEVENTS table. The preprocessing pipeline handles three critical tasks:

- 1) **Normalization:** Scaling raw VALUENUM data into a standardized range to stabilize model training.
- 2) **Log Transformation:** Applying logarithmic scaling to skewed distributions (e.g., lab results with extreme outliers) as specified in the project metadata.
- 3) **Metadata Mapping:** Utilizing a serialized metadata dictionary (.pkl) to ensure that all synthetic outputs can be accurately inverse-transformed back into clinically interpretable values.

II. LITERATURE REVIEW

Liu, Acharya, and Tan conducted a systematic review of deep learning approaches for synthetic healthcare data generation, examining methods published between 2019 and 2023 across Scopus, Web of Science, PubMed, and IEEE databases [2]. Their work provides the broadest contextual foundation for this study, establishing that GANs and VAEs demonstrated robust capabilities in generating realistic synthetic data for tabular, signal, image, and multi-modal datasets, while privacy-preserving approaches such as differential privacy and adversarial training significantly reduced re-identification risks

while maintaining data fidelity. Critically, the review highlights that challenges persist in preserving temporal correlations, reducing biases, and aligning with regulatory frameworks, particularly for longitudinal and high-dimensional data, challenges directly relevant to the ICU lab profile generation task undertaken in this work. The review further identifies that the primary issue lies in the assurance of privacy without undermining the utility of the generated synthetic data, and whether synthetic data should not still be considered personal data, an issue relevant to the comparative evaluation methodology employed here.

Wang, Myles, and Tucker propose a more operationally grounded framework for generating and evaluating cross-sectional synthetic electronic healthcare data, arguing that utility must be defined as clinical meaningfulness rather than statistical similarity alone [1]. Their framework evaluates synthetic data across four criteria: biological relationship preservation, univariate distance, multivariate distance, and privacy preservation, a structure that directly informed the evaluation design of this study. The univariate distance criterion, implemented via the Kolmogorov-Smirnov test, is explicitly validated in their work: a hypothesis test result, that is, p -value, for example, Kolmogorov-Smirnov test, can be used as a distance function for continuous variables providing direct methodological precedent for the KS statistic used as the primary evaluation metric across all 23 lab features in this study. Their multivariate framework further emphasizes that data with multiple dimensions often have correlated structure between data variables, and retaining such correlational structure in synthetic data is crucial to ensure that it is a truthful representation of the real world.

Wang et al. also surface a finding of direct relevance to the data quality challenges encountered in this work [1]. In their cardiovascular case study using the CPRD Aurum database, they demonstrate that the lifecycle of synthetic data production from ground truth selection to synthetic data generation, evaluation, and sensible synthetic data output must be fit into specific data generation methods according to different privacy requirements and that the ground truth selection process must explicitly address both privacy-sensitive variable identification and biological relationship definition before any generative modeling begins. This finding resonates strongly with the FLAG column filtering and double log-transformation issues encountered here, where upstream preprocessing decisions had greater impact on downstream model quality than any architectural choice, consistent with Wang et al.’s observation that additional consideration is required due to the longitudinal nature of healthcare data and long data format before using these approaches.

Together these works establish three pillars that contextualize the contributions of this study. First, that VAE-based approaches are theoretically well-suited to cross-sectional EHR generation and privacy preservation. Second, that KS statistics and correlation matrix analysis constitute a validated and widely applicable evaluation framework for synthetic healthcare data. Third, that data pipeline integrity is a prerequisite for

meaningful model comparison, a lesson that emerges empirically in this work and is implicitly supported by the reviewed papers’ emphasis on ground truth selection and preprocessing as first-order concerns rather than secondary steps.

III. GOALS

Given a patient’s past values of a specific lab test, how well can different generative and autoregressive models predict and generate the next lab value?

Analyze the accuracy of different generative models on LABEVENTS.

IV. DATA

Medical Information Mart for Intensive Care III (MIMIC-III) is a de-identified collection of data pertaining to several aspects of patient care and laboratory tests. Synthetic data was generated specifically for the LABEVENTS portion of the data containing clinical values for various lab tests.

A. Data Preprocessing

Preprocessing involved six major steps. Demographics were derived from PATIENTS and ADMISSIONS, computing age as the difference between admission time and date of birth in seconds to avoid integer overflow from MIMIC shifted dates, then capping at 90 to handle the anonymization of patients over 89. ICU stay features were extracted by taking the first ICU stay per admission sorted by admission time, mapping integer-encoded care unit codes back to unit names using the alphabetical cat.codes mapping, and bucketing length of stay into four clinically meaningful categories. Diagnosis features were encoded using a coarse CCS grouping of the primary ICD-9 code into 19 disease chapters, combined with 31 binary Elixhauser comorbidity flags computed from all secondary diagnoses. Lab values were extracted from LABEVENTS using a curated set of 23 ITEMIDs, filtered to a window of six hours before to 24 hours after ICU admission to capture baseline physiology, pivoted to wide format with one row per admission, clipped to physiologically plausible bounds, and imputed with population medians for genuinely missing values. Seven right-skewed labs (glucose, creatinine, lactate, PT, PTT, bilirubin, AST) were log-transformed using $\log_{10}$ after confirming they had not already been transformed upstream, with an inverse transform helper applying $\expm1$ at generation time. All features were then StandardScaler normalized with scalers fitted on the training split only to prevent data leakage.

B. cVAE

Both cVAE variants utilize the preprocessing technique above with a 85-15 train-val split.

C. ARM-LSTM

The Autoregressive Model (ARM) utilizes a StandardScaler and log-transformations for features with high skewness. Because this model treats patient data as a sequence rather than a static vector, it requires a specific sequential data restructuring phase. Each patient record is expanded into 23 distinct steps where the model learns clinical dependencies by predicting

$\tilde{x} = (x - \mu) / \sigma$, where μ and σ are the population mean and standard deviation fitted on the training set. The 60-dimensional conditioning vector c was similarly standardized. Missing lab values were imputed with per-lab population medians prior to scaling. At generation time, outputs are inverted in reverse order, first applying the StandardScaler inverse transform and then applying $\exp()$ for the log-transformed labs, to recover clinically interpretable units.

The model was optimized using Adam with a learning rate of 0.001 and a weight decay of 0.00001, the latter adding an L2 penalty equal to λ times the squared norm of the model parameters to the loss in order to discourage large weights. A ReduceLROnPlateau scheduler halved the learning rate whenever validation loss failed to improve for 5 consecutive epochs, preventing stagnation in flat loss regions. Gradient clipping with a maximum norm of 1.0 was applied at each update step, constraining the L2 norm of the gradient to remain at or below 1.0, to stabilize training against exploding gradients that commonly arise in sequential prediction tasks. Together, weight decay, dropout ($p=0.1$), and gradient clipping serve as the regularization strategy for the ARM.

C. PCARM

The PCARM builds on the autoregressive framework by using a probabilistic output at each step of generation instead of using a deterministic point estimate. The model will output the μ , the predicted mean, and the log variance, the predicted log uncertainty. During generation, a synthetic lab value will be drawn by sampling from the Gaussian distribution as $x = \mu + \sigma * \epsilon$. In this case, $\sigma = \exp(0.5 * \log_variance)$ and $\epsilon \sim \mathcal{N}(0,1)$. The log variance had a range of $[-6, 3]$ in order to prevent numerical instability from near zero estimates.

The architecture consists of a conditional encoder, a 2 layer LSTM, and a probabilistic output head. The condition encoder will map the 60 dimensional patient conditioning vector to a 128 dimensional embedding. This will then be used to initialize LSTM hidden and cell states before any labs can be generated. Throughout each step, the previous lab value and the encoded condition are merged and passed into the LSTM [9]. Its output is mapped to μ and log variance for the current lab. This leads to a total trainable parameters to be approximately 1,021,570.

Training uses teacher forcing with a START token of 0.0, the Gaussian negative log likelihood loss, the Adam optimizer at a learning rate $1e-3$, gradient clipping at norm 1.0, and an early stopping with a patience of 15. There were two modes of generation. The first one was a stochastic mode, where each value is sampled from the predicted distribution. The second one was a deterministic mode, where only the predicted mean was used. This allows a direct test if probabilistic sampling improves output diversity over point estimation.

D. GAN

Before training the model, the raw laboratory features from the dataset must be stabilized. This is done by passing the data through a PowerTransformer, mapping complex physiological variables into a normalized distribution. These critical preprocessing steps condition the data for the neural network to model effectively. Once the data is transformed, it is fed into the Generative Adversarial Network (GAN) architecture [10], which consists of two neural networks trained simultaneously in a zero-sum game. The Generator class takes in random noise and attempts to map it into a synthetic row of laboratory features that look realistically drawn from the preprocessed medical dataset. The Discriminator class acts as a binary classifier by taking rows of features and attempts to output a probability score indicating whether the sample is genuine. This is formulated around a standard minimax game where D tries to maximize its detection accuracy and G tries to minimize it.

Training GANs on tabular data is typically harder than the usual computer vision domains, especially on the medical dataset due to its unique structure. Real laboratory values rarely follow a clean, normal distribution meaning the standard Generator class has trouble making realistic data. To ensure a balanced optimization and counteract these issues, specific training constraints and structural strategies were implemented in this methodology. First, batch normalization layers were applied across the Generator to stabilize internal covariate shift during backpropagation, while spectral normalization bound the Discriminator's weights to prevent its gradients from exploding and completely overpowering the model. Finally, Leaky ReLU activation functions were used throughout both networks to mitigate the "dying neuron" problem by ensuring a small, non-zero gradient always flowed back through the layers, keeping the optimization active even during difficult training epochs.

When the generator successfully learns to output realistic data, these synthetic outputs still exist within the normalized, transformed scale. To make them usable for clinical and healthcare applications, they are passed through the inverse of the power transformer, reverting the synthetic features back to their original physiological ranges and true physical units.

E. RNN (Masked Lab Reconstruction)

The RNN is used for prediction, not generation. It does not create new synthetic patients. Instead, it takes a real patient's lab profile, hides a random subset of values, and learns to reconstruct the hidden ones from the remaining context. This is called masked reconstruction and is a predictive task, not a generative one. The distinction matters: the VAE and GAN output entirely new patient records, while the RNN outputs predicted values for specific positions in existing records [11].

The RNN serves two roles in the broader project. First, it is a standalone predictive model that answers the project question from a different angle than the generative models. Instead of synthesizing new patients, it tests whether inter-lab correlations can be learned and used for reconstruction.

Second, the RNN can be used as an evaluation tool for the VAE and GAN: by running the trained RNN on synthetic patients generated by those models and measuring reconstruction MAE, we can test whether the synthetic data preserved the real relationships between lab values. This gives a stronger evaluation of synthetic data quality than distributional metrics like KS statistic alone.

Model Architecture: The RNN is a 2-layer LSTM with hidden size 128, dropout of 0.4, and L2 weight decay of $1e-4$. At each sequence step, the model receives the masked lab value, a binary mask flag, and the 60-dimensional patient conditioning vector. The LSTM processes all 23 steps and a final linear layer outputs one scalar prediction per step.

TABLE I
RNN MODEL ARCHITECTURE

Component	Details
Input (per step)	Masked lab value + mask flag + 60-dim condition vector
LSTM Layer 1	Hidden size: 128, dropout: 0.4
LSTM Layer 2	Hidden size: 128, dropout: 0.4
Output Layer	Linear—one scalar value per sequence step
Loss	MSE on masked positions only

Masking Strategy: For each patient, 20 Loss Function: Loss is computed only on masked positions. This means the model is only penalized for values it had to guess, not the ones it could see.

The formula is: $\text{Loss} = \sum \text{mask}_i \cdot (y_i - \hat{y}_i) / \sum \text{mask}_i$

Baseline: A mean baseline was implemented for comparison. For each masked lab position, the baseline predicts the training set mean for that specific lab across all training patients. This requires no learning and sets a minimum bar: if the RNN cannot beat always guessing the average, it has not learned anything useful. The baseline is computed separately per lab, so glucose gets the glucose training mean, hematocrit gets the hematocrit training mean, and so on. *Training:* The model was trained using the Adam optimizer with a learning rate of 0.001 and weight decay of $1e-4$. Early stopping was used with a patience of 7 epochs, meaning training stops if validation loss does not improve for 7 consecutive epochs and the best weights are restored. The train/validation split was 80/20. *Improvements:* The initial model was trained without weight decay and with dropout of 0.2. This resulted in significant overfitting. The training loss reached 0.36 while validation loss was 0.86, a gap of 0.50. The following changes were made: Dropout increased from 0.2 to 0.4 L2 weight decay of $1e-4$ added to Adam optimizer Early stopping added with patience of 7 epochs These changes reduced the train/validation gap from 0.50 to 0.09 and improved overall MAE from 8.37 to 7.73.

VI. RESULTS

A. ARM

The ARM generated 49,304 synthetic patient records conditioned on the same demographic, ICU, and diagnostic features used by the cVAE. Across all 23 lab values, the synthetic data exhibited severe mean regression, collapsing individual patient

TABLE II
KS STATISTIC TABLE

Feature	ARM LSTM	cVAE	SWD-cVAE	GAN	PCARM
glucose	1.0000	0.184	0.063	0.1499	0.048
sodium	0.4923	0.105	0.090	0.2163	0.078
potassium	0.4572	0.178	0.069	0.1275	0.089
creatinine	0.9708	0.165	0.145	0.1828	0.111
bun	0.5522	0.129	0.134	0.0978	0.070
bicarbonate	0.4901	0.153	0.076	0.3700	0.111
chloride	0.4749	0.099	0.049	0.3778	0.068
calcium_total	0.5486	0.113	0.095	0.2236	0.107
free_calcium	0.7528	0.299	0.326	0.5126	0.304
phosphate	0.4606	0.155	0.108	0.1708	0.133
ph	0.6845	0.218	0.185	0.2485	0.204
pco2	0.6023	0.231	0.244	0.3686	0.203
po2	0.6884	0.241	0.185	0.2056	0.285
lactate	0.9973	0.221	0.268	0.3739	0.278
hemoglobin	0.4589	0.068	0.042	0.1709	0.018
hematocrit	0.4499	0.063	0.041	0.2176	0.018
wbc	0.4715	0.150	0.074	0.1282	0.137
platelets	0.4688	0.122	0.069	0.2232	0.069
pt	1.0000	0.141	0.098	0.1100	0.138
ptt	1.0000	0.117	0.093	0.0621	0.115
bilirubin	0.9670	0.296	0.371	0.2949	0.347
ast	1.0000	0.284	0.289	0.4504	0.347
albumin	0.6767	0.338	0.335	0.3675	0.349

variability into a narrow spike around the population mean for every lab. This is quantified by a mean standard deviation ratio of 0.686, meaning synthetic lab values had only 68.6 percent of the spread seen in real data. The Kolmogorov–Smirnov test rejected distributional similarity for all 23 labs (p approximately 0.00 in every case), and Wasserstein distances were large across the board, with glucose reaching 127.06 and po2 reaching 86.33. The correlation heatmap made this failure most visually striking: real data showed a rich structure of positive and negative inter-lab correlations, while the ARM’s synthetic data produced a near-uniformly red matrix, indicating spurious near-perfect correlations driven entirely by all patients receiving similar predicted values rather than genuine physiological relationships. The Frobenius distance between real and synthetic correlation matrices was 21.98.

These results reflect fundamental limitations of the autoregressive architecture when applied to tabular clinical data rather than implementation failures. The ARM’s sequential design assumes that each lab value can be meaningfully predicted from those preceding it in the sequence, implying that glucose informs sodium, which informs potassium, and so on. In reality, first-day ICU lab values are jointly determined by the patient’s underlying physiological state rather than causally linked in sequence, making this assumption biologically invalid. Furthermore, the gap between teacher-forced training and autoregressive inference, known as exposure bias, compounded errors across all 23 steps, with later labs such as PT and PTT achieving KS statistics of 1.0, indicating complete distributional separation from real data.

B. GAN

The Generative Adversarial Network (GAN) generated 49,304 synthetic patient records, maintaining sample size consistency with the other baseline models evaluated in this study.

The model achieved its highest fidelity and best performance when replicating ‘ptt’ and ‘bun’ values, closely matching the true clinical distributions of the training data. These strong results were primarily driven by the initial data transformation stage. Passing the highly skewed, multimodal physiological data through a power transformer successfully stabilized the features into a Gaussian-like distribution before training, allowing the Generator to map continuous random noise to these complex clinical feature spaces far more effectively. Additionally, the integration of structural training constraints, specifically batch normalization across the Generator and spectral normalization to bound the Discriminator’s weights, successfully prevented the models from overpowering one another. This balanced optimization mitigated the common tabular pitfall of mode collapse, ensuring that the final synthetic cohort accurately preserved the structural variations and independent correlations of the original medical dataset.

C. PCARM

The PCARM generated 49,304 synthetic patient records that were conditioned on the same demographic, ICU, and diagnostic features used by the other models. This achieved a mean KS statistic of 0.153.

The model had its best results in hemoglobin and hematocrit with a KS statistic of 0.018. The model also performed quite well across other areas such as glucose, chloride, BUN and platelets, with 0.048, 0.068, 0.070, and 0.069, respectively. However, performance did see a decline with other lab values. Bilirubin, AST, and albumin all ranged from 0.347-0.349. It is worth noting that these all fall under the liver panel and this performance decline pattern can be seen across all of the models presented. This presents the possibility that can be attributed to extreme distributional skew rather than model specific limitations.

Comparing the stochastic and deterministic generation did confirm the benefit of doing probabilistic sampling. The stochastic variant produced standard deviation ratios around 1.0 across most of the labs. This can indicate that the sampling from the predicted distributions may partially recover the variance lost by point estimation. This can address the mean regression that can be seen in a deterministic ARM, which had a mean standard deviation of 0.686. This directly supports the core design hypothesis of the PCARM.

D. cVAE

Following correction of the FLAG filtering error and the double log-transformation issue, both models produced substantially improved results over earlier runs. The WAE-SWD variant outperformed the KL variant across the majority of evaluation criteria.

Training behavior: The WAE-SWD showed significantly more stable training curves. Train and validation losses tracked identically throughout all 300 epochs, completely resolving the val-below-train anomaly seen in the KL variant. The SWD regularization term dropped from 0.40 to 0.04 within the first

20 epochs and remained stable. The KL variant exhibited larger cycle-reset spikes and a persistent generalization gap.

Distribution fidelity: WAE-SWD produced dramatically better lab distributions and correlation between lab values than KL. The collapsed spikes at median values that characterized the KL variant were absent. Glucose extended to 1400, BUN extended beyond 100, and bicarbonate showed a broad spread from 10 to 50. All improvements directly attributable to the aggregate rather than per-patient regularization. Potassium’s bimodal structure at peaks 3.0 and 4-5 was reproduced, which was entirely absent in the KL variant.

KS statistics: The Kolmogorov-Smirnov test comparing real and synthetic distributions showed WAE-SWD improved 17 of 23 labs relative to KL. Good labs (KS $\leq$ 0.10) increased from 3 to 12. POOR labs (KS $\geq$ 0.20) decreased from 8 to 6. Notable improvements include glucose (0.184 to 0.063), potassium (0.178 to 0.069), bicarbonate (0.153 to 0.076), and WBC (0.150 to 0.074). Six labs regressed, most notably bilirubin (0.296 to 0.371) and lactate (0.221 to 0.268), likely due to SWD aggregate regularization smoothing extreme tail values for these high-variance labs.

TABLE III
RNN RESULTS

Metric	Value
Labs	23
Mask Ratio	20%
Overall MAE (RNN)	7.73
Overall MAE (Baseline)	10.10
Labs where RNN beats baseline	20/23
Hematocrit MAE	2.597
Train/val gap	0.09

E. RNN

The RNN outperformed the mean baseline on 20 of 23 labs with an overall MAE of 7.73 compared to 10.10 for the baseline, a 23% improvement. Hematocrit had an MAE of 2.597 against a clinical range of 5 to 70. The initial model without regularization overfit noticeably, with a train/validation loss gap of 0.50. Adding dropout, weight decay, and early stopping reduced this to 0.09 and improved overall MAE from 8.37 to 7.73. High-variance labs like po2 and platelets were still hard to predict even though the RNN beat the baseline on both.

VII. CONCLUSION

A. Future Work

Several extensions of the RNN approach would strengthen the project. First, the trained RNN could be applied directly to synthetic patients generated by the VAE and GAN to evaluate whether those models preserved real inter-lab correlations. If reconstruction MAE on synthetic patients is similar to MAE on real patients, it provides stronger evidence of synthetic data quality than KS statistics alone. Second, the RNN could be retrained on a combined dataset of real and synthetic patients to test whether VAE or GAN output is useful for data augmentation. If MAE improves with synthetic data added, it

directly validates the generative models for downstream use. Third, replacing the one-snapshot-per-patient structure with true time-ordered lab readings from raw LABEVENTS would allow the RNN to model how lab values change over time, which is clinically more meaningful and would likely improve performance on high-variance labs like po2 and platelets that all models currently struggle with.

B. Summary and Lessons Learned

Again, we report that WAE-SWD and PCARM emerge as the strongest performing architectures with mean KS statistics of 0.163 and 0.164 respectively, with WAE-SWD demonstrating superior electrolyte and blood gas fidelity while PCARM achieves the best haematology results by exploiting known inter-variable correlations through autoregressive conditioning.

The main takeaway from this work aside from model performance is that correcting two upstream preprocessing errors, a FLAG column filtering issue that removed 65% of normal lab results and a double log-transformation affecting multiple labs, improved distributional fidelity more substantially than any model architecture decision. Six labs (free calcium, lactate, bilirubin, AST, albumin, pO2) remain poorly reproduced across all five architectures, with consistent failure attributable to upstream-unknown data transformations and extreme distributional skew rather than model limitations, suggesting that targeted data strategies rather than architectural innovation are required to address these remaining challenges.

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